

AH-LoRA: Evaluating Resource-Adaptive Decentralized LoRA Against Pooled Training

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Abstract

Can clients with unequal computing budgets collaboratively train low-rank adapters while retaining the accuracy of ordinary LoRA trained on pooled data? We evaluate AH-LoRA, which combines capability-constrained rank selection, domain-dependent contribution weights, and neighbor-to-neighbor aggregation of scaled LoRA updates. A final peer reduction constructs a rank-16 deployment model. Nine protocols are compared on all 50,000 CIFAR-100 training examples and 10,000 test examples, using frozen ResNet-18 features, 15 clients, and three seeds. Pooled LoRA reaches $57.23 \pm 0.59\%$ full-test accuracy, whereas the combined adaptive-rank, domain-weighted method reaches $6.90 \pm 2.03\%$. A substantial gap already exists without rank adaptation: centralized federated aggregation reaches $20.06 \pm 0.44\%$ and uniform-rank peer training reaches $12.69 \pm 3.55\%$. Within the domain-weighted comparison, adaptation loses 3.89 ± 1.44 percentage points against fixed feasible ranks. It reduces training-factor payload by 41.9%, but total deployment payload falls only 0.26% once the implemented dense control exchange and final assembly are included. These results do not meet the accuracy-preservation objective under the tested 30-round protocol; they do not establish an impossibility result for decentralized LoRA. Raw examples are absent from simulated peer messages, but shared updates and training statistics have no demonstrated privacy protection.

1 Introduction

LoRA adapts a pretrained model by training low-rank matrices while keeping the original weights fixed [2]. Federated learning permits participants to retain their examples locally [4]; decentralized optimization replaces a dedicated aggregation server with communication between participating peers [1, 3]. Together these ideas suggest a practical objective: match ordinary pooled-data LoRA accuracy while allowing participants with smaller computing budgets to train smaller adapters.

Accuracy preservation is the criterion in this study. An improvement over pooled training is not required. Conversely, matching another weak federated baseline is insufficient if both methods trail a model trained directly on the pooled data. The proposed benefit also has two distinct parts: keeping raw examples local is a property of the training workflow, whereas protecting information in exchanged updates requires a separate privacy mechanism and threat model. A possible hospital collaboration motivates the problem but is not an evaluated application.

Combining the components introduces difficulties that isolated experiments do not resolve. LoRA factors are not identifiable, heterogeneous ranks induce different scaling and compression, and nonuniform contribution weights must survive the gossip normalization. The final model also matters: an adapter assembled centrally by an evaluator is not necessarily a model produced by the stated peer protocol.

We therefore evaluate a concrete composition. AH-LoRA uses scaled effective-update aggregation, a source-pinned adaptive-rank controller, the canonical conservative domain-weighting function, a

weighted Metropolis kernel, and an explicit final peer reduction. We compare its output with a conventional pooled LoRA model under common data, initialization, and evaluation. Supporting experiments on oracle hierarchy, automatic discovery, and isolated rank and weighting policies are reported separately. This is an empirical study of the implemented method, not a claim of a new convergence theorem or a validated private-learning system.

2 Learning Objective and Evaluation

Let $\mathcal{D}_i$ contain n_i training examples at client i , let $n = \sum_i n_i$, and let $\phi(x) \in \mathbb{R}^{512}$ be the shared frozen feature extractor. A client’s logits are

$$z_i(x) = (W_0 + \Delta W_i)\phi(x) + b_0, \quad \Delta W_i = \frac{\alpha}{r_i} B_i A_i, \quad 1 \leq r_i \leq R_i^{\max}. \quad (1)$$

The frozen head (W_0, b_0) , frozen features, and LoRA initialization are shared within each seed. Local loss is the empirical cross entropy

$$f_i(\Delta W) = \frac{1}{n_i} \sum_{(x,y) \in \mathcal{D}_i} \ell((W_0 + \Delta W)\phi(x) + b_0, y). \quad (2)$$

Ordinary pooled LoRA trains one rank- R adapter directly on all examples:

$$\min_{A,B} F_{\text{pool}}(A, B) = \sum_i \frac{n_i}{n} f_i\left(\frac{\alpha}{R} B A\right), \quad R = 16. \quad (3)$$

Sample-weighted federated comparators target the same empirical weighting. Quality and domain arms instead use round-dependent weights $\pi_i^t \propto n_i q_i^t \lambda_i^t$. These weights change the training emphasis; all methods are still evaluated on the same unweighted full test set.

The primary endpoint is the final assembled rank-16 model’s accuracy on all 10,000 test examples. For seed s and method m , we report the paired difference in percentage points,

$$\Delta_{m,s} = 100 \left(\text{Acc}(\widehat{\Delta W}_{m,s}; \mathcal{T}) - \text{Acc}(\Delta W_{\text{pool},s}; \mathcal{T}) \right). \quad (4)$$

A prespecified noninferiority margin was not available; we report observed differences and sample standard deviations without equating a nonsignificant difference with equivalence. Personalized accuracy, $N^{-1} \sum_i \text{Acc}(\Delta W_i; \mathcal{T}_i)$, is a secondary diagnostic. It evaluates different models on different local test shards and cannot replace the pooled-model comparison.

3 Method

3.1 Capability-constrained adaptive rank

The integrated experiment uses the Project 1 controller’s decision rule rather than a replacement loss-threshold heuristic. Capability ceilings are $R_i^{\max} \in \{4, 8, 16\}$ and the feasible menu is

$$\mathcal{R}_i = \{r \in \{2, 4, 6, 8, 12, 16, 24, 32\} : r \leq R_i^{\max}\}. \quad (5)$$

A temporary probe at the client’s ceiling measures gradient stable rank before local training. For each nonzero two-dimensional LoRA factor gradient G , define $s(G) = \|G\|_F^2 / \|G\|_2^2$; the client signal s_i^t is the median over these gradients. The probe uses one fixed, seeded local training minibatch, independently of the training shuffle. No test labels enter the controller.

The smoothed demand and candidate target are

$$d_i^t = \text{clip}(s_i^t, 0, R_i^{\max}), \quad \bar{d}_i^t = 0.7\bar{d}_i^{t-1} + 0.3d_i^t, \quad (6)$$

$$r_i^* = \text{nearest}_{\mathcal{R}_i} \left(\max \{2, 0.5c_i R_i^{\max}, \min(\bar{d}_i^t, R_i^{\max})\} \right), \quad (7)$$

where $c_i \in \{0, 1/2, 1\}$ identifies the capability tier; the first demand observation initializes the moving average. Ties in the candidate quantizer choose the smaller rank, subject to its capability floor. The deployed policy starts at the ceiling, holds it for two warm-up rounds, and imposes a minimum feasible rank of at least $R_i^{\max}/2$. An upward request requires $r_i^* > r_i$ and $\bar{d}_i^t \geq 1.10r_i$; a downward request requires $r_i^* < r_i$ and $\bar{d}_i^t \leq 0.85r_i$. Two consecutive requests in the same direction move one menu step. A post-training quality score, defined below, falling more than 5% below its preceding moving average forces restoration to the ceiling at the next decision. Quality is smoothed with weights 0.8/0.2. Optional residual-demand modulation in the generic controller is disabled here.

A rank increase from r to r' preserves the current update by scaling each existing factor by $\sqrt{r'/r}$. New A rows are randomly initialized and their corresponding B columns are zero, preserving the output while allowing the added directions to learn. A decrease uses the truncated effective-update SVD. Gradient probes and quality evaluations are logged separately from training; a smaller training rank does not by itself establish a reduction in total computation. The controller is source-pinned to Project 1; its one-minibatch probe is an explicit adaptation of Project 1's multi-minibatch measurement procedure to this benchmark.

3.2 Quality and domain contribution weights

After local optimization, client i evaluates cross entropy ℓ_i^t on its fixed training probe and sets

$$q_i^t = (1 + \ell_i^t)^{-1}, \quad p_i^t = \frac{n_i q_i^t}{\sum_j n_j q_j^t}. \quad (8)$$

Both sides of the fixed/adaptive comparison use this same quality definition. Domain weighting uses five Project 2 signals. Let n_{ic} be training class counts, $v_{ic} = n_{ic}/n_i$, $g_c = \sum_i n_{ic}/n$, and let

$$u_i^t = \text{vec}(\Delta W_i^{t,\text{local}} - \Delta W_i^{t,\text{pre}}), \quad \bar{u}^t = N^{-1} \sum_i u_i^t. \quad (9)$$

The feature vector comprises Jensen–Shannon divergence to the pooled training histogram, Euclidean and cosine distance to the mean update, normalized class entropy, and class imbalance:

$$\begin{aligned} x_{i1} &= \text{JS}(v_i, g), & x_{i2} &= \|u_i - \bar{u}\|_2, & x_{i3} &= 1 - \frac{u_i^\top \bar{u}}{\|u_i\|_2 \|\bar{u}\|_2}, \\ x_{i4} &= -\frac{\sum_{c: v_{ic} > 0} v_{ic} \log v_{ic}}{\log C}, & x_{i5} &= \frac{\max_c n_{ic}}{\min_{c: n_{ic} > 0} n_{ic}} \left(1 + \frac{\#\{c : n_{ic} = 0\}}{C} \right). \end{aligned} \quad (10)$$

Cosine distance is zero when its denominator is at most 10^{-12} . For distributions a, b and $m = (a+b)/2$, $\text{JS}(a, b) = \frac{1}{2} \text{KL}(a\|m) + \frac{1}{2} \text{KL}(b\|m)$ and $\text{KL}(a\|b) = \sum_c a_c \log(a_c/b_c)$. The implementation adds 10^{-12} and renormalizes at each divergence call.

The conservative allocator directly imports the canonical Project 2 function. Each feature is standardized across clients and clipped to $[-3, 3]$, with a constant feature contributing zero. With

$K = 5$, its score and bounded multiplier are

$$\begin{aligned} z_{ik} &= \text{clip}\left(\frac{x_{ik} - \bar{x}_k}{\sigma_k}, -3, 3\right), & a_i &= K^{-1} \sum_k z_{ik}, \\ e_i &= \frac{\exp(\text{clip}(a_i, -4, 4))}{\sum_j p_j \exp(\text{clip}(a_j, -4, 4))}, & h_i &= 1 + 0.10(e_i - 1), \end{aligned} \quad (11)$$

$$\lambda_i = \text{clip}(ch_i, 0.85, 1.15), \quad \sum_i p_i \lambda_i = 1. \quad (12)$$

Bisection determines the positive scalar c . The resulting gossip target is

$$\pi_i = \frac{n_i q_i \lambda_i}{\sum_j n_j q_j \lambda_j}. \quad (13)$$

Quality-only arms set $\lambda_i = 1$; sample-only arms set both $q_i = \lambda_i = 1$. Domain weighting is tested for its effect on the parity objective, without requiring it to improve accuracy over quality-only training.

3.3 Effective-update aggregation and weighted gossip

LoRA factors can be gauge transformed as $(B, A) \mapsto (BQ, Q^{-1}A)$ for any invertible Q . This leaves BA unchanged but generally changes factor averages. AH-LoRA therefore sends the factors and reconstructs $(\alpha/r)BA$ at the receiver. The shared α and source rank are used in every forward pass and merge.

For the undirected training graph, form the symmetric Metropolis proposal

$$Q_{ij} = \frac{1}{1 + \max(\deg(i), \deg(j))} \quad \text{on edges}, \quad Q_{ii} = 1 - \sum_{j \neq i} Q_{ij}. \quad (14)$$

To retain nonuniform contribution weights, the actual kernel is

$$P_{ij}^t = Q_{ij} \min\left(1, \frac{\pi_j^t}{\pi_i^t}\right) \quad (i \neq j), \quad P_{ii}^t = 1 - \sum_{j \neq i} P_{ij}^t. \quad (15)$$

It is row stochastic, preserves graph support, and satisfies detailed balance:

$$\pi_i P_{ij} = Q_{ij} \min(\pi_i, \pi_j) = \pi_j P_{ji}, \quad \pi^\top P = \pi^\top. \quad (16)$$

It is generally neither symmetric nor doubly stochastic. A separate validator checks these weighted identities; the uniform-mean convergence claims of the earlier symmetric runner are not applied to this kernel.

After local training, each client forms and compresses

$$Y_i^t = \sum_j P_{ij}^t \Delta W_j^{t, \text{local}}, \quad \Delta W_i^{t+1} = \mathcal{C}_{r_i^t}(Y_i^t). \quad (17)$$

If $Y = U \Sigma V^\top$, the stored factors are

$$B = \frac{U_{:r} \Sigma_{:r}^{1/2}}{\sqrt{\alpha/r}}, \quad A = \frac{\Sigma_{:r}^{1/2} V_{:r}^\top}{\sqrt{\alpha/r}}, \quad \mathcal{C}_r(Y) = U_{:r} \Sigma_{:r} V_{:r}^\top. \quad (18)$$

This is the minimum-Frobenius-error rank- r approximation, with discarded energy $\|Y - \mathcal{C}_r(Y)\|_F^2 = \sum_{k>r} \sigma_k(Y)^2$. Before compression, the weighted mean is preserved for the current π . After compression its deviation is exactly $\sum_i \pi_i [\mathcal{C}_{r_i}(Y_i) - Y_i]$. Since local optimization, ranks, and weights change over time, these algebraic identities do not imply convergence or accuracy parity. The optional error-feedback extension stores $m_i^{t+1} = Y_i^t + m_i^t - \mathcal{C}_{r_i}(Y_i^t + m_i^t)$; it is evaluated only in the separate historical ablation.

3.4 Peer exchange of weighting information

The domain policy requires global training statistics. These inputs are explicitly delivered over the graph rather than supplied through an uncharged evaluator callback. Each client provides (n_i, q_i) ; domain arms additionally provide its 100-bin training histogram and 51,200-element effective-update change. A breadth-first spanning tree gathers these records to a participating root peer and broadcasts the gathered records over the same tree. Each peer independently computes Equations (10)–(13) from its received view, and equality of the resulting weights is checked.

This straightforward allgather is intentionally fully accounted for. If record i has L_i numeric fields and the gather subtree at peer v is S_v , its gather cost is $8 \sum_{i \in S_v} (L_i + 1)$ bytes, including one source identifier per record. Each broadcast edge carries $8 \sum_i (L_i + 1)$ bytes. Thus

$$B_{\text{control}}^t = 8 \sum_{v \neq \text{root}} \sum_{i \in S_v} (L_i + 1) + 8(N - 1) \sum_i (L_i + 1). \quad (19)$$

The numerical transport model uses eight-byte scalars and identifiers. The simulator shares read-only array references internally but charges full logical transmissions. This exchange reveals all contributed records to every peer; it is not secure aggregation.

3.5 Final assembly and resource accounting

A final model is produced by the peer protocol itself. Before testing, we choose the first client with the largest capability as deployment root and use a spanning tree of the training graph. Let $\text{ch}(i)$ denote tree children. Each peer accumulates a dense weighted numerator and scalar mass,

$$H_i = \pi_i \Delta W_i + \sum_{j \in \text{ch}(i)} H_j, \quad M_i = \pi_i + \sum_{j \in \text{ch}(i)} M_j. \quad (20)$$

Every non-root sends (H_i, M_i) to its parent. Only the root refactorizes:

$$\widehat{\Delta W} = \mathcal{C}_{16}(H_{\text{root}}/M_{\text{root}}). \quad (21)$$

There is no intermediate truncation in this reduction. It constructs the weighted mean before the final rank projection using $N - 1$ existing graph edges. It requires dense temporary storage and a root capable of the final SVD. The root is a participating peer, not an external training server; failure tolerance and root rotation are not evaluated.

For a source adapter with rank r_j , the training payload is $S_j = r_j(512 + 100)$ fp32 scalars. Let $E_t = \{(i, j) : i \neq j, P_{ij}^t > 0\}$. The deployment payload accounting is

$$B_{\text{total}} = 4 \sum_{t=1}^T \sum_{(i,j) \in E_t} S_j^t + \sum_{t=1}^T B_{\text{control}}^t + (N - 1)(4 \cdot 51200 + 8). \quad (22)$$

Table 1: Corrected comparison. All final deployment models have rank 16. Quality and domain arms form a 2×2 ablation of rank policy and domain factors.

Arm	Training rank	Contribution weights	Training coordination
Pooled	16	individual samples	direct pooled Adam
Pooled-reset	16	individual samples	pooled Adam, reset
FedAvg-16	16 per client	n_i	central FedAvg
MH-16	16 per client	n_i	weighted peer MH
MH-sample	fixed 4/8/16	n_i	weighted peer MH
Fixed-quality	fixed 4/8/16	$n_i q_i$	weighted peer MH
Fixed-domain	fixed 4/8/16	$n_i q_i \lambda_i$	weighted peer MH
Adaptive-quality	P1 policy	$n_i q_i$	weighted peer MH
Adaptive-domain	P1 policy	$n_i q_i \lambda_i$	weighted peer MH

The last term is one final peer assembly. Earlier assemblies used to draw evaluation curves are recorded separately and do not alter training. FedAvg already returns its centrally aggregated model and does not require the peer reduction; pooled training has no inter-client payload. Static sample-count exchange, graph setup, packet framing, and subsequent model dissemination are excluded. We report training rank-sample products, $\sum_{t,i} r_i^t n_i$, and probe work separately; this proxy excludes SVD, feature extraction, optimizer, and communication costs and is not a measured total-FLOP or hardware-speedup claim.

4 Experimental Design

4.1 Matched pooled and integrated comparison

The primary benchmark uses all 50,000 CIFAR-100 training images and 10,000 test images. Frozen ImageNet-pretrained ResNet-18 features have dimension 512 after deterministic resizing to 224×224 and ImageNet normalization. Only LoRA factors on the random frozen 100-class head are trained. This is head-only LoRA on cached features, not end-to-end backbone or language-model fine-tuning.

Five groups of CIFAR-100 superclasses define the domain partition. Each domain has three clients, split by Dirichlet(0.5). The training sets and test shards are disjoint, and their unions are checked against the entire dataset. Heterogeneous arms assign repeating ceilings (4, 8, 16); uniform controls use rank 16 everywhere. A seeded permutation independent of domain ordering defines the ring, and the same graph is used by paired arms. Known domains determine the benchmark partition but do not select a privileged training hierarchy in this experiment.

All nine arms use $\alpha = 32$, learning rate 10^{-3} , weight decay 10^{-4} , minibatch size 128, one local epoch per round, 30 rounds, and seeds 42, 43, and 44. Table 1 specifies the comparators. Ordinary pooled training preserves Adam moments across epochs. Pooled-reset is an additional control that resets Adam each epoch; federated clients reset it each communication round. This separates the conventional baseline from the optimizer-reset convention used by the federation.

Each arm processes 1.5 million training-example exposures, excluding separately logged probes. Equal exposure does not mean equal optimizer steps: pooled training performs $\lceil 50000/128 \rceil$ steps per epoch, whereas the federation performs $\sum_i \lceil n_i/128 \rceil$. Counts are recorded rather than assumed equal. The pooled rank-16 model is the conventional reference, while the fixed heterogeneous arms isolate adaptation under the same client capability ceilings. They are different controls and answer different questions.

Hyperparameters, method list, seeds, and the final round are fixed before the full comparison.

Test evaluation produces curves but is not used for rank decisions, weighting, early stopping, or model selection. Reported uncertainties are sample standard deviations across three seeds, not confidence intervals. Rank-policy effects are paired as adaptive minus fixed within each weighting condition; domain effects are paired as domain minus quality within each rank policy. Source hashes, feature-cache identity, initialization hashes, split manifests, phase timings, ranks, weights, matrices, transport ledgers, and final adapters are retained.

4.2 Implementation validation and excluded pilots

Validation checks scaled heterogeneous merging, weighted stationarity and detailed balance, graph-restricted traffic, trainable rank expansion, deterministic training probes, and final peer assembly against a direct weighted-SVD test oracle. Paired smoke runs check that adaptive probes leave fixed/adaptive warm-up trajectories unchanged and that pooled and pooled-reset have identical first-epoch behavior. The direct weighted average is used to validate the reduction, not to replace it at evaluation.

An audit of earlier pilot results found issues that affect their interpretation. First, the old weighting wrapper computed Sinkhorn(DWD) for an already doubly-stochastic W and positive diagonal D . Since diagonal rescaling is precisely what Sinkhorn balancing removes,

$$\text{Sinkhorn}(DWD) = W \tag{23}$$

under the usual uniqueness conditions, up to numerical tolerance. Consequently, identical weighted and unweighted pilot outcomes do not establish that domain weighting is ineffective or preserves accuracy: the intended intervention was canceled.

Second, the pilot’s so-called adaptive-rank arm used a median-loss threshold and half-ceiling rule, not the Project 1 gradient-based controller. Its measured performance describes that heuristic only. Third, the separate pooled pilot hardcoded $\alpha = 32$ while the compared decentralized run used $\alpha = 16$; it also contrasted a sample-based pooled objective with client-uniform gossip and an evaluator-assembled consensus adapter. These are not matched tests of Equation (4). We retain their raw artifacts for provenance but exclude their success/failure and weighting-neutrality conclusions. The corrected experiment uses one common scaling constant, the intended policies, explicit objective controls, and actual peer assembly.

4.3 Supporting component studies

The earlier Project 3 protocol battery used 50 rounds, the same three seeds, and uniform rank 16 or heterogeneous ranks (4, 12, 32). It compared client-uniform local, FedAvg, flat gossip, and an oracle hierarchy. This objective, rank schedule, and round budget differ from the corrected comparison. Historical consensus scores use a centrally computed uniform mean followed by a reference-rank SVD; they are diagnostic scores, not outputs of the new final tree reduction.

The oracle uses known domains and the matrix $W^t = W_{\text{intra}} W_{\text{bridge}} W_{\text{intra}}$ on bridge rounds, with intra-domain mixing otherwise. Its separate online-discovery extension projects effective updates into 64 dimensions, normalizes them, and smooths them:

$$z_i^t = \frac{P \text{vec}(\Delta W_i^t)}{\|P \text{vec}(\Delta W_i^t)\|_2}, \quad h_i^t = \frac{0.8h_i^{t-1} + 0.2z_i^t}{\|0.8h_i^{t-1} + 0.2z_i^t\|_2}. \tag{24}$$

With affinity $a_{ij} = h_i^\top h_j$, a graph-supported kernel $K_{ij} = \mathbf{1}_{(i,j) \in E \cup I} \exp[(a_{ij} - a_{\max})/\tau]$ is Sinkhorn-balanced and blended with a self-weight floor. This pairwise-affinity kernel differs from the canceled

Arm	Full-test accuracy (%)	Paired gap to pooled (pp)	Seeds
Pooled rank 16	57.23 ± 0.59	$+0.00 \pm 0.00$	3
Pooled rank 16; Adam reset	57.25 ± 0.66	$+0.03 \pm 0.07$	3
FedAvg rank 16	20.06 ± 0.44	-37.17 ± 0.40	3
Peer rank 16	12.69 ± 3.55	-44.54 ± 4.12	3
Peer ranks 4/8/16; sample weights	7.71 ± 0.81	-49.51 ± 1.38	3
Fixed ranks; quality weights	10.69 ± 0.73	-46.54 ± 0.47	3
Fixed ranks; domain weights	10.78 ± 2.01	-46.44 ± 2.08	3
Adaptive ranks; quality weights	6.48 ± 1.67	-50.75 ± 2.08	3
Adaptive ranks; domain weights	6.90 ± 2.03	-50.33 ± 2.46	3

Table 2: Corrected integrated comparison on the full CIFAR-100 test set. Each entry is the mean $\pm$ sample standard deviation across seeds. Differences are calculated within matching seeds against pooled rank-16 LoRA with persistent Adam moments. Peer results evaluate the final rank-16 assembly. The backbone is frozen; peer communication is simulated in one process. No noninferiority or equivalence margin was prespecified.

Paired contrast	Accuracy difference (pp)
Adaptive minus fixed; quality only	-4.21 ± 1.79
Adaptive minus fixed; domain weighted	-3.89 ± 1.44
Domain minus quality; fixed ranks	$+0.10 \pm 1.63$
Domain minus quality; adaptive ranks	$+0.42 \pm 0.38$
Rank $\times$ domain interaction	$+0.32 \pm 1.35$

Table 3: Paired factorial contrasts, mean $\pm$ sample standard deviation across seeds. Positive values favor the first condition. All four arms use the same graph and sample-times-quality base allocation. The interaction is $(A_D - F_D) - (A_Q - F_Q)$. These are descriptive contrasts, not equivalence tests.

clientwise diagonal weighting above. Cluster count is chosen by maximum silhouette over $k \in \{2, \dots, \min(8, N - 1)\}$; nonpositive scores select one group. Labels are used only to score discovery. The historical callback observes all adapters centrally, so this discovery result is not a neighborhood-local protocol and is not included in the corrected fixed-ring ablation.

Project 1’s original five-task battery compared a controller against fixed rank 32, above its capability ceilings. A later Fashion-MNIST rerun used a feasible fixed (4, 8, 16) reference and the revised controller. Project 2 separately evaluated recorded leave-one-client-out contributions, $y_i = \text{Acc}(\text{all}) - \text{Acc}(\text{all} \setminus i)$, on 75 real-data observations. Its exploratory fitted score used ridge regression, $\hat{\theta}_\kappa = (X^\top X + \kappa I)^{-1} X^\top y$, with leave-one-task-out selection of κ . The corrected training experiment uses the conservative unfitted allocator in Equation (12), not a fit to held-out test accuracy.

5 Results

5.1 Corrected pooled comparison

The corrected experiment does not meet the pooled-accuracy criterion (Table 2). Ordinary pooled LoRA reaches $57.23 \pm 0.59\%$, compared with $6.90 \pm 2.03\%$ for the complete adaptive-domain arm: a seed-paired difference of -50.33 ± 2.46 percentage points. Every federated arm trails the pooled reference in every seed. This is a measured negative result for the specified implementation and training budget, not an inference from the previously mismatched pilots.

The shortfall is not confined to adaptive rank. Uniform-rank FedAvg reaches $20.06 \pm 0.44\%$,

uniform-rank peer MH reaches $12.69 \pm 3.55\%$, and heterogeneous sample-weighted MH reaches $7.71 \pm 0.81\%$. Pooled-reset remains close to ordinary pooled training at $57.25 \pm 0.66\%$, with a paired difference of $+0.03 \pm 0.07$ points. Thus resetting Adam in pooled training does not reproduce the much larger federated shortfall, although this control does not isolate optimizer behavior under local domain shifts and repeated factor refactorization.

The factorial comparison isolates the selected policies more directly (Table 3). Adaptive-quality reaches $6.48 \pm 1.67\%$ versus $10.69 \pm 0.73\%$ for fixed-quality, a paired change of -4.21 ± 1.79 points. With domain weights, the corresponding change is -3.89 ± 1.44 points against fixed-domain’s $10.78 \pm 2.01\%$. Domain factors change accuracy by $+0.10 \pm 1.63$ points at fixed ranks and $+0.42 \pm 0.38$ points under adaptive ranks. Unlike the canceled pilot operation, this weighting intervention changes the actual mixing matrices and trajectories. Its small observed effects do not restore pooled accuracy; a claim of broad no-regression is also unwarranted from three seeds, particularly given the variability at fixed ranks. No improvement over conventional training was required for success.

Rank adaptation reduces mean training-factor payload from 20.56 MB to 11.96 MB in quality-only training, and to 11.95 MB with domain weighting, approximately 41.9% in either condition. Including the required control traffic and final assembly, quality-only deployment payload falls from 23.62 MB to 15.01 MB, a 36.4% reduction accompanied by the measured accuracy loss. Domain-control exchange alone costs 3.275 GB per run; its total deployment payload is 3.299 GB at fixed ranks and 3.290 GB with adaptation, only a 0.26% reduction. MB and GB here denote 10^6 and 10^9 bytes. Each adaptive run additionally processes 57,600 gradient-probe examples and 57,600 quality-probe examples. These costs preclude interpreting the factor reduction as a total-compute saving.

All 27 runs process the same 1.5 million training-example exposures. They record 11,730 pooled Adam steps and 12,000 aggregate client Adam steps, with the latter determined by client minibatch boundaries. Independent evaluation of every saved checkpoint reproduces all 27 recorded fp32 correct counts on all 10,000 examples, and verifies rank-16 final updates and prediction hashes. A separate fp64 factorized forward pass disagrees on four near-tied predictions across 270,000 evaluations. Only one changes a correct count: seed-43 adaptive-quality increases by one of 10,000 examples, or 0.01 percentage points, with a top-two fp64 logit margin of 5.96×10^{-8} . The original fp32 results and the initial strict cross-precision check are retained. This precision sensitivity does not change the reported conclusion.

5.2 What the component evidence establishes

Table 4 retains the earlier 50-round protocol results. With uniform rank 16, the oracle hierarchy reaches $68.98 \pm 0.95\%$ personalized accuracy, while its diagnostic consensus accuracy is $10.25 \pm 0.62\%$. This contrast illustrates why a domain-specific personalized score cannot establish parity of one globally deployed model. The oracle result remains useful evidence that known domain structure changes the behavior of local models.

Under the historical heterogeneous schedule, factor zero-padding gives personalized accuracy of $4.71 \pm 0.22\%$ for FedAvg, $20.63 \pm 4.40\%$ for MH, and $31.24 \pm 1.27\%$ for the oracle hierarchy, compared with $44.07 \pm 0.70\%$ for effective-update oracle mixing. Error feedback gives $26.63 \pm 1.68\%$ for MH and $41.28 \pm 0.71\%$ for the oracle. These are finite-run ablations of representation and compression, not universal performance orderings.

The separate online-discovery experiment selects five groups with adjusted Rand index 1.00 at stage 10 for each seed. At stage 20 its scores are 1.00, 0.918, and 1.00, with the middle seed selecting six groups. This demonstrates recovery under coordinator-visible observations; it neither establishes a privacy property nor substitutes for the fixed-ring pooled comparison.

The original Project 1 controller stayed at rank 2 throughout its five-round battery and reduced

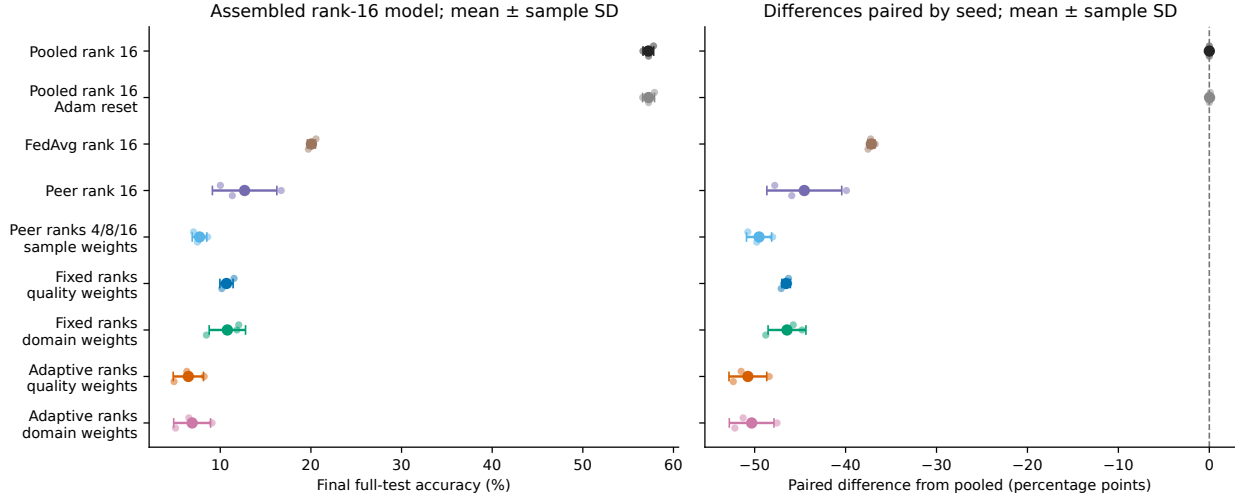


Figure 1: Corrected final rank-16 full-test comparison. All methods use the same test set; error bars show sample standard deviation across seeds 42–44. Ordinary pooled LoRA is the primary reference.

Table 4: Supporting historical Project 3 results after 50 rounds, in percent (mean $\pm$ sample standard deviation, seeds 42–44). The objective is client-uniform; historical consensus is an evaluator-assembled uniform average. These values are not the corrected pooled comparison.

Rank context	Method	Personalized	Diagnostic consensus
Uniform 16	Local	57.93 ± 0.55	14.23 ± 1.27
	FedAvg	23.20 ± 0.29	23.20 ± 0.29
	MH	42.30 ± 1.39	15.14 ± 5.10
	Oracle hierarchy	68.98 ± 0.95	10.25 ± 0.62
Heterogeneous 4/12/32	Local	53.60 ± 0.62	14.50 ± 0.80
	FedAvg	5.74 ± 0.71	5.73 ± 0.59
	MH	28.45 ± 2.42	6.83 ± 0.45
	Oracle hierarchy	44.07 ± 0.70	11.47 ± 1.13

the reported training-FLOP estimate by 93.75% against fixed rank 32, while losing accuracy on three of five tasks. Since rank 32 violates every tested client ceiling, that comparison does not measure a feasible hardware-matched accuracy trade-off. The revised Fashion-MNIST rerun finishes at 82.85% for both the adaptive controller and feasible fixed ranks (4, 8, 16), with a reported 10.0% reduction in the training-FLOP estimate. This is one task, one seed, and five rounds; its unscaled adapter convention also differs from the corrected scaled-head benchmark.

Project 2’s conservative follow-up changes Spearman association with recorded contributions from 0.152 to 0.196 and pairwise ordering accuracy from 0.520 to 0.547. These are model-free ranking results, not retrained-model accuracy results. Its earlier ridge forms have weak global fit ($R^2 = 0.079$ and 0.025) and task-dependent correlations. Neither the isolated Fashion-MNIST parity nor the contribution-ranking change predicts the composed method’s full-test accuracy without the corrected experiment.

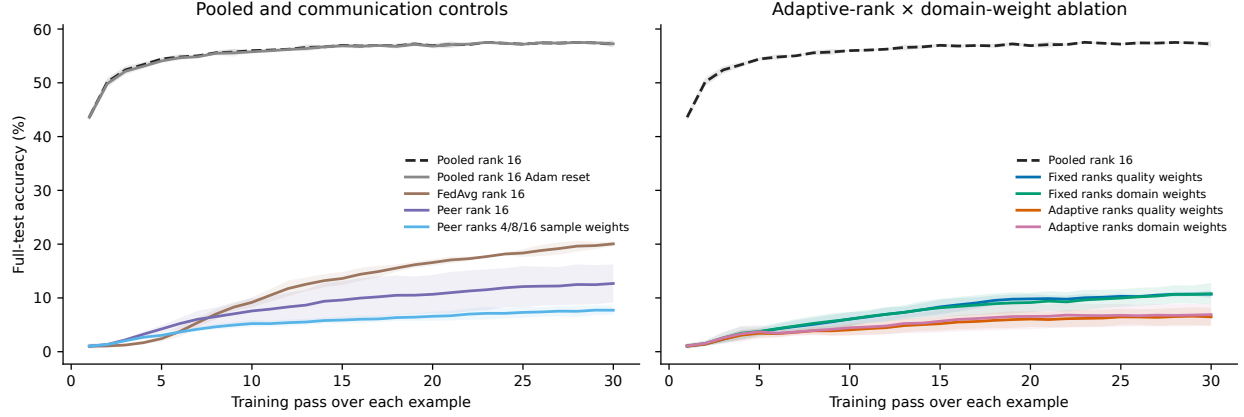


Figure 2: Full-test accuracy over the fixed 30-round budget. Each curve evaluates the actual deployment model at that round. Intermediate peer assembly supports evaluation only and does not feed a global model back into decentralized training.

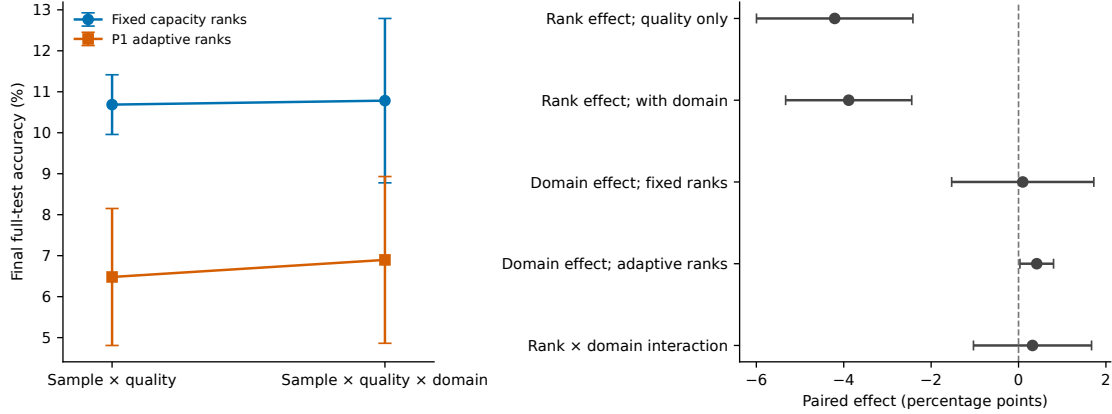


Figure 3: Paired component effects within the corrected experiment. Rank adaptation is compared at the same weighting condition; domain factors are compared at the same rank policy. These comparisons complement the primary comparison against pooled training.

6 Discussion and Limitations

The parity question has several distinct comparisons. Pooled versus FedAvg tests the effect of local optimization and aggregation with uniform rank. MH-16 adds neighbor communication. MH-sample imposes heterogeneous capacity. The four quality/domain arms then isolate the selected adaptive policy and the domain multipliers. A shortfall at one step cannot automatically be attributed to a different component, and matching a resource-constrained federated control does not establish matching pooled training.

The diagnostics support a cautious interpretation of the large gap. FedAvg’s mean per-round relative discarded energy is only 1.39×10^{-6} across the three runs. The final adaptive-domain assembly discards a mean relative energy of 7.63×10^{-5} . A large final Frobenius-energy loss therefore is not an adequate explanation by itself. The uniform-rank peer arm also reaches $42.10 \pm 1.42\%$ personalized accuracy while its full-test assembled model reaches $12.69 \pm 3.55\%$, illustrating the distinction between local specialization and a globally useful deployment model. Local domain heterogeneity

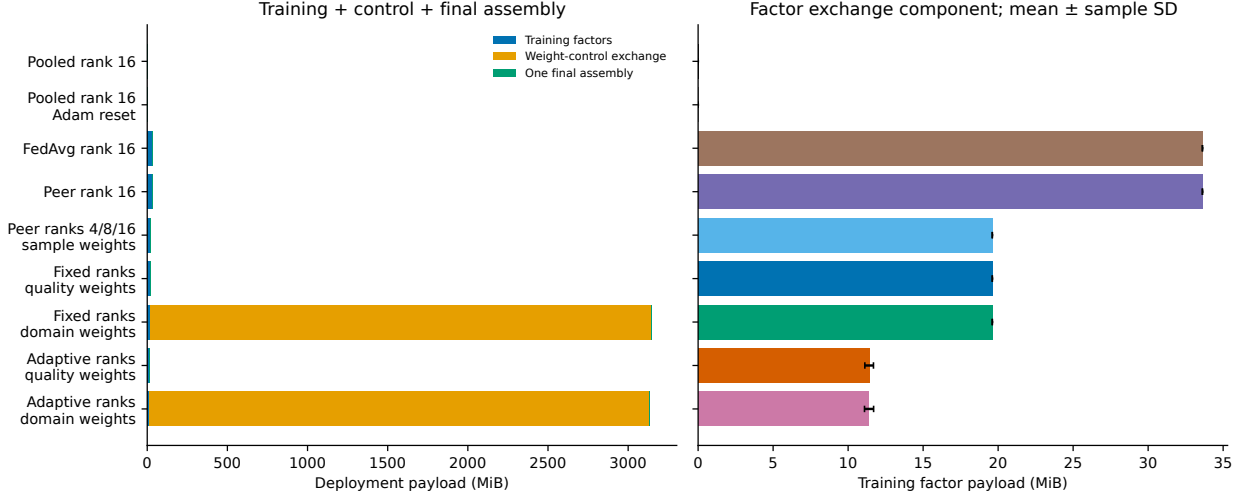


Figure 4: Simulated deployment payload: factor exchange during training, all weighting-control traffic, and one final peer assembly. Dense control information is included. These counts are not measured bandwidth or network latency.

and the optimization trajectory are plausible contributors, but their causal contributions are not separately identified by these measurements.

Equal global sample exposures, even with all optimizer steps counted, do not make local-then-average training equivalent to pooled sequential optimization. Local objectives differ, Adam’s gradient transformation depends nonlinearly on its moments, and SVD refactorization changes factor coordinates before the next optimizer reset. In general, if $\mathcal{U}_i$ is a local optimization map and $\mathcal{U}_{\text{pool}}$ its pooled counterpart,

$$\mathcal{C}_R \left(\sum_i \pi_i \mathcal{U}_i(\Delta W) \right) \neq \mathcal{U}_{\text{pool}}(\Delta W). \quad (25)$$

This explains why the desired equality is an empirical hypothesis rather than an algebraic property of weighted assembly. Small Frobenius truncation energy does not rule out losing predictive directions or changing later optimization. The ablation measures the specified 30-round protocol; it does not causally isolate every difference between local and pooled optimization, prove an impossibility result for federated LoRA, or justify extending training until a preferred test result appears.

The benchmark deliberately exposes costs that a factor-only account omits. Domain weighting currently exchanges dense update changes and class histograms with every peer, and final exact assembly also uses dense matrices. This simple protocol makes data dependencies auditable but can dominate the savings from smaller factor payloads. A compressed sufficient-statistic protocol would be a different implementation requiring its own correctness and accuracy evaluation. Rank-sample products likewise omit gradient probes, factorization, optimizer work, and dense processing.

The execution is a deterministic single-process simulation of peer message flow on the designated remote GPU machine. It verifies graph-constrained operations and payloads; it does not establish multi-host reliability, measured communication speed, asynchronous behavior, or tolerance of dropped clients. The cached-feature, head-only CIFAR-100 task does not validate end-to-end visual fine-tuning, language models, medical data, or arbitrary AI training. Three seeds describe variation in this testbed and do not establish broad equivalence. The historical oracle and automatic-discovery results have different observation assumptions and are not silently included in the corrected pipeline.

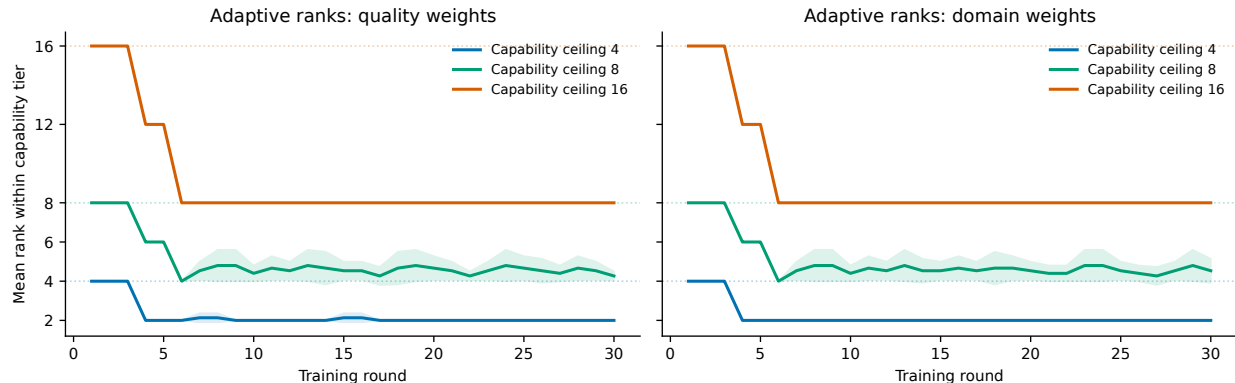


Figure 5: Adaptive ranks under the published capability ceilings. Rank choices and gradient probes are recorded for each client; rank reductions alone do not account for probe, SVD, or metadata costs.

Raw examples are absent from the simulated peer messages, and each client’s optimizer reads its assigned examples. The simulator process nevertheless has access to the entire dataset, so this experiment does not verify enforced isolation between machines. Updates and training statistics are exposed; in particular, the domain-control allgather reveals all supplied records to all peers. No differential privacy, secure aggregation, membership-inference evaluation, reconstruction evaluation, or medical threat model is implemented in this study. Consequently, accuracy parity, if observed, would still not justify a claim that hospitals can collaborate without privacy loss.

7 Conclusion

The completed experiment does not achieve the original accuracy-preservation objective. Under the corrected 30-round protocol, pooled rank-16 LoRA reaches $57.23 \pm 0.59\%$ full-test accuracy and the combined adaptive-domain method reaches $6.90 \pm 2.03\%$. The gap already appears in fixed-rank federated controls, and the tested adaptive policy introduces a further loss against capability-matched fixed ranks. Domain weighting has small observed effects and does not recover pooled performance. Smaller adapters reduce factor traffic, but dense domain-control exchange largely eliminates the reduction in total payload. These findings describe the tested framework and budget, not an impossibility result for decentralized federated LoRA. The implementation and its independent verification establish an executable, auditable experiment; they establish neither accuracy parity nor a privacy guarantee for the proposed hospital use case.

A Reproducibility and Supporting Figures

The repository contains executable protocols, canonical policy provenance, feature and split hashes, per-round JSON records, phase timings, final adapters, and scripts that generate the reported tables and plots. The corrected driver is `project-3-hierarchical-gossip/experiments/integrated_benchmark.py`. Its evaluation does not tune on test data. Remote tests cover the numerical operators and paired execution invariants; artifacts distinguish full experiments from smoke tests and excluded pilots. Local credentials are not part of the artifact bundle. Historical figures below retain their original protocols and should be read with the associated qualifications.

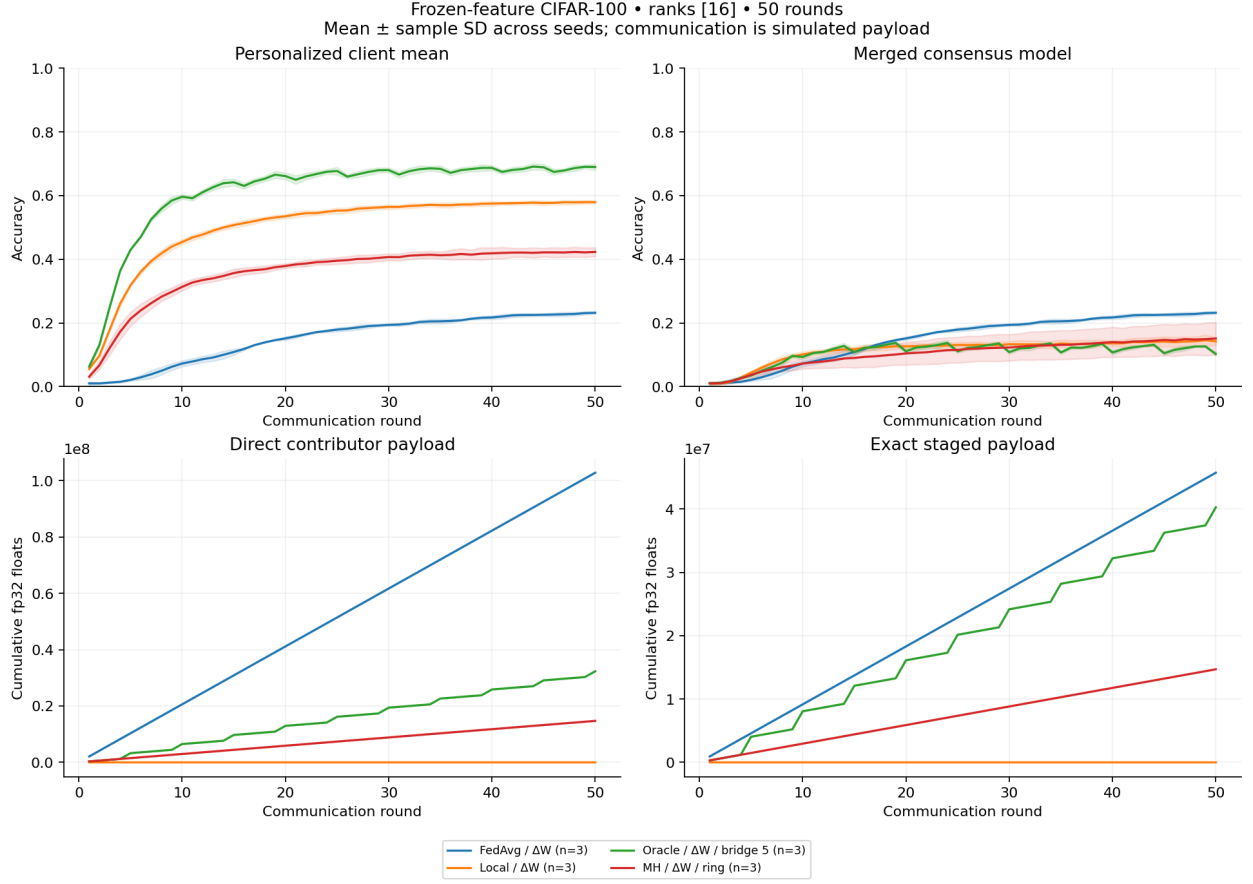


Figure 6: Supporting historical uniform-rank protocol curves, 50 rounds. Personalized and evaluator-assembled consensus accuracy are different endpoints.

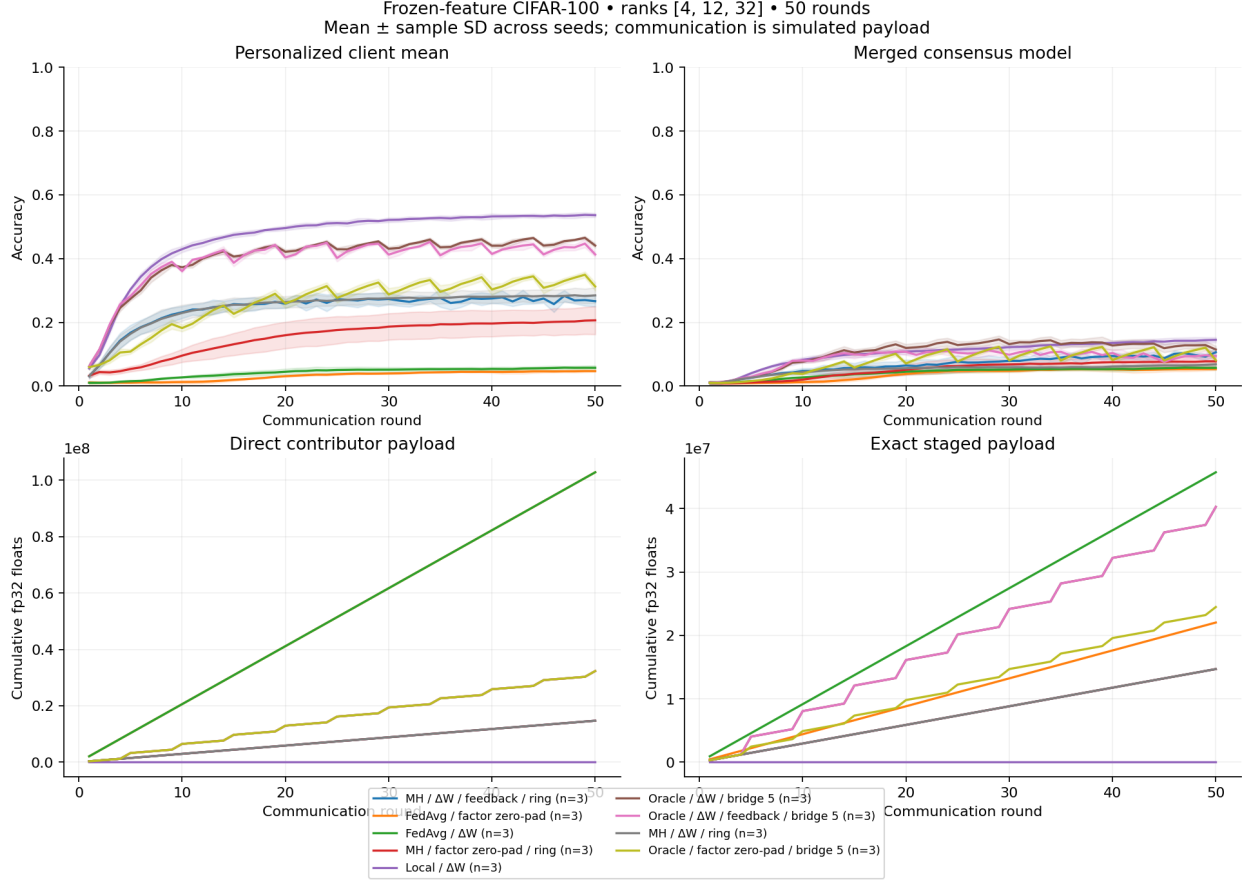


Figure 7: Supporting historical heterogeneous (4, 12, 32) protocol curves. The rank budget and objective differ from the corrected (4, 8, 16) pooled comparison.

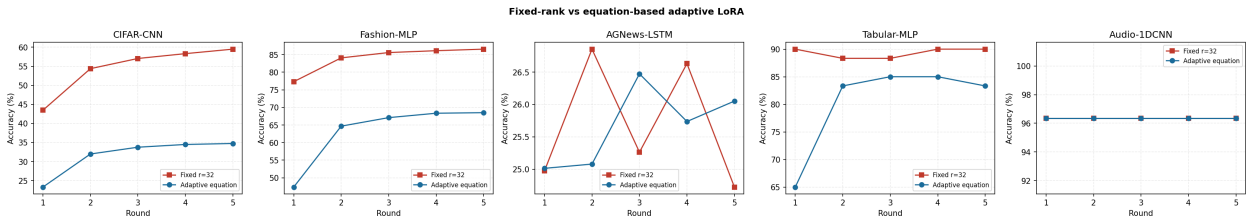


Figure 8: Historical five-task Project 1 battery. Fixed rank 32 exceeds the tested capability ceilings and is not a feasible hardware-matched reference. The revised Fashion-MNIST comparison is reported separately.

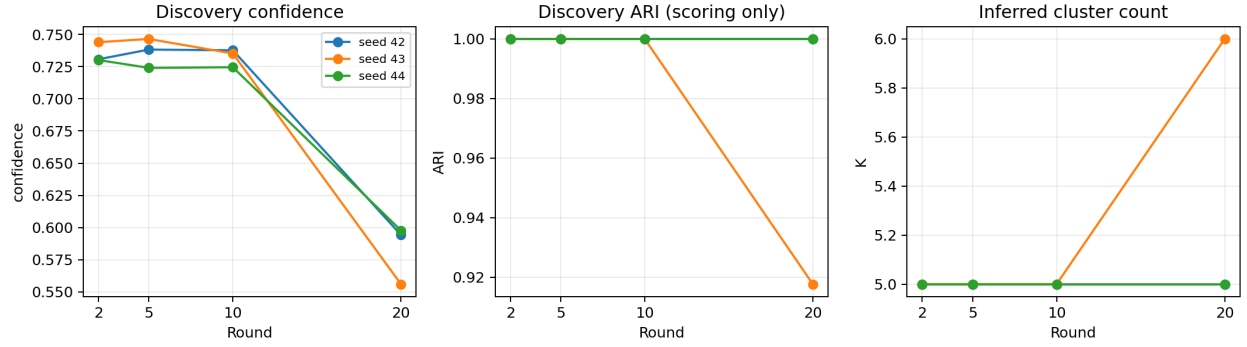


Figure 9: Supporting online-discovery results under coordinator-visible adapter observations. This experiment evaluates group recovery and is separate from the corrected fixed-ring pipeline.

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